

Seminar Project Progress Report – II

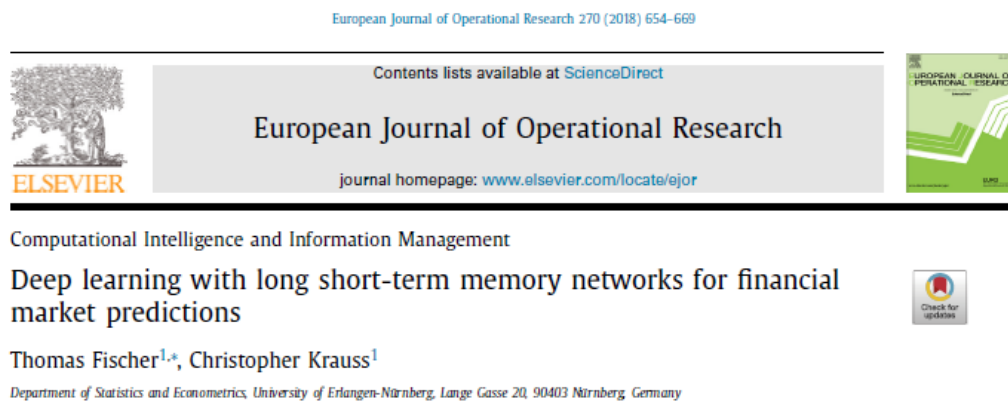
Implement baseline model

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Problem Statement

1. Implement Fischer & Krauss (2018), "Deep learning with LSTM networks for financial market predictions"
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1 Introduction

Financial time-series prediction is an inherently challenging task due to the highly noisy, non-stationary, and dynamic nature of financial markets. Stock price movements are influenced by numerous economic, political, and behavioral factors, making accurate forecasting difficult. Traditional statistical methods often struggle to capture complex temporal dependencies present in financial data.

The paper extends the work of Krauss et al. (2017) by investigating whether Long Short-Term Memory (LSTM) networks can improve financial market prediction and portfolio construction compared to conventional machine learning approaches.

For a fair comparison with previous studies, the authors employ the same dataset and benchmark models. The study considers stocks from the S&P 500 index over the period from January 1990 to October 2015.

1.1 Motivation

Previous studies reported that conventional machine learning models achieved promising results in financial prediction tasks. For example:

- Logistic Regression served as a simple baseline classifier.
- Random Forest achieved average daily returns of approximately 0.43%.
- Deep Neural Networks (DNNs) achieved average daily returns of approximately 0.33%.

However, Deep Neural Networks were found to be highly sensitive to training configurations and hyperparameter settings, often leading to unstable performance. Since financial data possess strong temporal dependencies, the authors proposed the use of LSTM networks, which are specifically designed to model sequential and time-dependent information.

1.2 Main Contributions of the Paper

The paper makes three major contributions:

(a) **LSTM-based Financial Prediction Framework**

The study develops and evaluates an LSTM-based framework for predicting whether a stock will outperform or underperform the cross-sectional median return on the next trading day.

(b) **Interpretation of the LSTM Black Box**

The authors analyze the characteristics of stocks selected by the LSTM model and identify common patterns among profitable trades, thereby providing insights into the model's decision-making process.

(c) **Construction of a Simplified Reversal Strategy**

Based on the patterns learned by the LSTM, the authors design a transparent short-term reversal trading strategy and compare its profitability with the LSTM-based portfolio.

The study compares the performance of the following models:

- Logistic Regression (LOG),
- Random Forest (RAF),
- Deep Neural Network (DNN),
- Long Short-Term Memory Network (LSTM).

1.3 Dataset Requirements and Availability

The original study by Fischer and Krauss (2018) uses a comprehensive financial dataset obtained from Thomson Reuters Datastream.

1.3.1 Required Dataset Components

The following information is required to reproduce the original study.

1. Stock Universe

- S&P 500 constituent stocks.
- Historical index components from December 1989 to October 2015.
- Delisted companies should be included whenever possible to avoid survivorship bias.

2. Daily Stock Data For each stock, the following variables are required:

- Date
- Ticker
- Open Price
- High Price
- Low Price
- Close Price
- Total Return Index / Total Return Price (dividend and split-adjusted)
- Trading Volume

3. Company Information

- Company Name
- GICS Sector
- GICS Industry
- Historical Market Capitalization

4. Corporate Actions

- Stock Splits
- Dividends
- Delisting Information

5. Market Benchmark Data For the S&P 500 Index (SPX):

- Date
- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume

6. Valuation Variables

- Price-to-Earnings Ratio (P/E)
- Price-to-Book Ratio (P/B)
- Dividend Yield

7. Risk Factor Data

- Market Excess Return (Market-RF)
- SMB (Small Minus Big)
- HML (High Minus Low)
- Momentum (MOM)
- Short-Term Reversal
- Long-Term Reversal
- Risk-Free Rate (RF)

1.3.2 Dataset Available for Current Implementation

Due to the unavailability of Thomson Reuters Datastream data at present, Yahoo Finance (`yfinance`) is being used as a fallback data source for initial implementation and pipeline development.

Table 1: Comparison Between Original Dataset Requirements and Available Data

Dataset Component	Available	Remarks
Historical S&P 500 Constituents	No	Only current constituents available from Wikipedia/Yahoo Finance
Delisted Companies	No	May introduce survivorship bias
OHLCV Stock Data	Yes	Available through Yahoo Finance
Dividend-Adjusted Prices	Yes	Using <code>auto_adjust=True</code> as proxy
True Total Return Index	No	Not identical to Thomson Reuters Total Return Index
Company Name	Yes	Available
Current GICS Sector/Industry	Yes	Available from Wikipedia
Historical GICS Classification	No	Not available
Historical Market Capitalization	No	Not available from free data source
Stock Splits	Yes	Available through Yahoo Finance
Dividends	Yes	Available through Yahoo Finance
Delisting Information	No	Not available
S&P 500 Benchmark Data	Yes	Downloaded using <code>gSPC</code>
P/E Ratio History	No	Not available as historical daily series
Price-to-Book Ratio History	No	Not available as historical daily series
Dividend Yield History	No	Not available as historical daily series
Fama-French Factors	Not Yet	Can be downloaded from Kenneth French Data Library
Risk-Free Rate	Not Yet	Can be downloaded from Kenneth French Data Library

2 Methodology

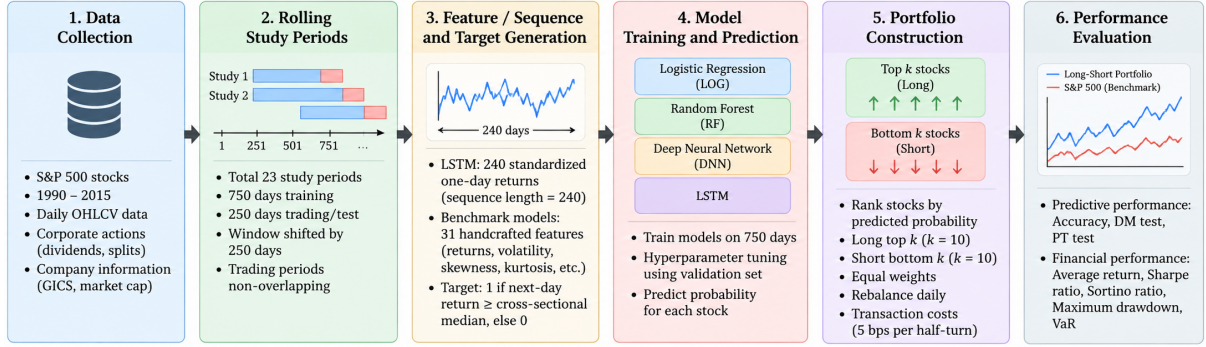


Figure 1: Overall methodology followed in this study.

2.1 Rolling Study Period Framework

Following Fischer and Krauss (2018), the complete dataset spanning January 1990 to October 2015 is divided into 23 rolling study periods. Each study period consists of a total of 1000 trading days.

- Training Period: First 750 trading days
- Trading/Test Period: Last 250 trading days

The model is trained using information available during the first 750 trading days and is subsequently evaluated on the following 250 trading days in a completely out-of-sample manner.

After completion of one study period, the entire 1000-day window is shifted forward by 250 trading days. Consequently, the training windows overlap across consecutive study periods, whereas the 250-day trading periods remain non-overlapping.

Table 2: Illustration of Rolling Study Periods

Study Period	Training Window	Trading/Test Window
1	Days 1–750	Days 751–1000
2	Days 251–1000	Days 1001–1250
3	Days 501–1250	Days 1251–1500

This design ensures a realistic backtesting framework in which future information never enters the training process.

2.2 Construction of Training and Trading Sets

For study period i , let n_i denote the number of stocks that are members of the S&P 500 index at the final day of the corresponding training period.

Only stocks belonging to the index at that point in time are included in both the training and trading sets. Since index constituents change over time, the value of n_i may vary slightly across study periods.

A stock is not required to possess a complete 750-day history. Newly listed firms are included using all historical observations available up to the end of the training period.

Similarly, during the trading period, if a stock becomes unavailable due to delisting, acquisition, merger, bankruptcy, or any other corporate event, it remains tradable until its final available trading day. This procedure reduces survivorship bias and better reflects real-world market conditions.

2.3 Feature Generation for LSTM

Let $P_t^{(s)}$ denote the adjusted closing price of stock s at time t .

The simple return over m trading days is defined as

$$R_t^{(m,s)} = \frac{P_t^{(s)}}{P_{t-m}^{(s)}} - 1. \quad (1)$$

For the LSTM model, only the standardized one-day return is used:

$$\tilde{R}_t^{(1,s)} = \frac{R_t^{(1,s)} - \mu_{train}}{\sigma_{train}}, \quad (2)$$

where μ_{train} and σ_{train} are estimated exclusively from the training period to avoid look-ahead bias.

A sequence length of 240 trading days is employed, corresponding approximately to one trading year.

Therefore, for stock s , an input sequence is constructed as

$$X_t^{(s)} = \{\tilde{R}_{t-239}^{(1,s)}, \tilde{R}_{t-238}^{(1,s)}, \dots, \tilde{R}_t^{(1,s)}\}. \quad (3)$$

Sequences are generated using a rolling-window approach.

The first training sample is

$$[1, 240] \rightarrow 241,$$

the second sample is

$$[2, 241] \rightarrow 242,$$

and the process continues until the end of the training period.

For a training window of 750 trading days and sequence length 240, each stock contributes approximately

$$750 - 240 = 510$$

training sequences.

2.4 Target Construction

The prediction problem is formulated as a binary classification task.

For every trading day $t + 1$, the one-day return of all available stocks is computed and ranked cross-sectionally.

The cross-sectional median return is then determined.

The target variable is defined as

$$Y_{t+1}^{(s)} = \begin{cases} 1, & \text{if } R_{t+1}^{(1,s)} \geq \text{Median}(R_{t+1}) \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Thus,

- Class 1 represents stocks expected to outperform the median stock.
- Class 0 represents stocks expected to underperform the median stock.

This construction ensures balanced target classes and aligns with the methodology adopted in the original paper.

2.5 Model Training

Four predictive models are considered:

- (a) Logistic Regression (Baseline)
- (b) Random Forest
- (c) Deep Neural Network (DNN)
- (d) Long Short-Term Memory Network (LSTM)

For Logistic Regression, Random Forest, and DNN, the input consists of 31 handcrafted return-based features proposed by Krauss et al. (2017).

For the LSTM model, the input consists of a sequence of 240 standardized one-day returns.

All models are trained using only the information available during the corresponding training period.

3 Forecasting, Ranking and Trading Strategy

After training, the model predicts the probability that a stock will outperform the cross-sectional median return on the next trading day. Following Fischer and Krauss (2018), all portfolio decisions are based on these predicted probabilities.

3.1 Step 1: Forecasting Next-Day Outperformance Probability

For every stock s on day t , the trained model estimates

$$\hat{P}_{t+1|t}^s = P(Y_{t+1}^s = 1 \mid \text{Information available up to time } t) \quad (5)$$

where

$$Y_{t+1}^s = \begin{cases} 1, & \text{if stock } s \text{ outperforms the cross-sectional median return at } t+1 \\ 0, & \text{otherwise} \end{cases}$$

Thus, the model does not directly predict the future return. Instead, it predicts the probability that a stock will belong to the upper half of the return distribution on the following day.

3.2 Step 2: Probability Estimation from the Output Layer

The final layer of the classifier consists of two output neurons followed by a Softmax activation function.

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{j=1}^2 e^{z_j}} \quad (6)$$

Example:

$$\text{Output} = [0.08, 0.92]$$

which implies

$$P(Y = 0) = 0.08$$

and

$$P(Y = 1) = 0.92$$

Therefore, the model estimates a 92% probability that the stock will outperform the cross-sectional median return on the next trading day.

3.3 Step 3: Ranking Stocks

For each trading day, all stocks are sorted according to their predicted probability

$$\hat{P}_{t+1|t}^s$$

in descending order.

Stock	Predicted Probability
MSFT	0.99
AAPL	0.51

Although both stocks belong to Class 1, MSFT receives a stronger signal because its predicted probability is substantially higher.

Therefore, ranking preserves the strength of the prediction signal, whereas simple classification would treat both stocks identically.

3.4 Why Ranking is Preferred over Classification

A classification rule would purchase all stocks predicted as Class 1 and sell all stocks predicted as Class 0.

However, the probability values contain additional information about prediction confidence.

For example,

$$P(Y = 1) = 0.99$$

represents a much stronger signal than

$$P(Y = 1) = 0.51.$$

Consequently, the paper ranks stocks according to predicted probabilities and trades only the strongest signals.

3.5 Step 4: Long–Short Portfolio Construction

After ranking all stocks, a long–short portfolio is formed.

For a chosen value of k ,

- Long Position: Top- k stocks with highest probabilities.
- Short Position: Bottom- k stocks with lowest probabilities.

The paper evaluates

$$k \in \{10, 50, 100, 150, 200\}.$$

The primary portfolio reported in the paper uses

$$k = 10.$$

Hence,

- Top 10 stocks $\rightarrow$ Long Portfolio
- Bottom 10 stocks $\rightarrow$ Short Portfolio

resulting in a portfolio containing

$$2k = 20$$

stocks.

3.6 Step 5: Equal Weighting Scheme

Following Huck (2009, 2010), equal weighting is applied.

For the long portfolio,

$$w_i = \frac{1}{k}$$

for each stock.

Similarly, for the short portfolio,

$$w_i = \frac{1}{k}.$$

Example for $k = 10$:

- Total Long Capital = \$100
- Invest \$10 in each of the 10 long stocks
- Total Short Capital = \$100
- Short \$10 in each of the 10 short stocks

3.7 Step 6: Dollar-Neutral Portfolio

The portfolio is constructed to be dollar neutral.

$$\text{Long Exposure} = +100\%$$

$$\text{Short Exposure} = -100\%$$

Therefore,

$$\text{Net Exposure} = 0.$$

A dollar-neutral strategy attempts to remove broad market movements and isolate stock-selection skill.

3.8 Step 7: Portfolio Return Calculation

Let

$$R_{\text{Long}}$$

denote the equally weighted average return of the long portfolio and

$$R_{\text{Short}}$$

the equally weighted average return of the short portfolio.

The daily long-short portfolio return is

$$R_P = R_{\text{Long}} - R_{\text{Short}} \tag{7}$$

Example:

$$R_{\text{Long}} = 1.5\%$$

$$R_{\text{Short}} = -1.0\%$$

Then,

$$R_P = 1.5\% - (-1.0\%) = 2.5\%.$$

3.9 Overall Forecasting and Trading Pipeline

The complete workflow used in this study is

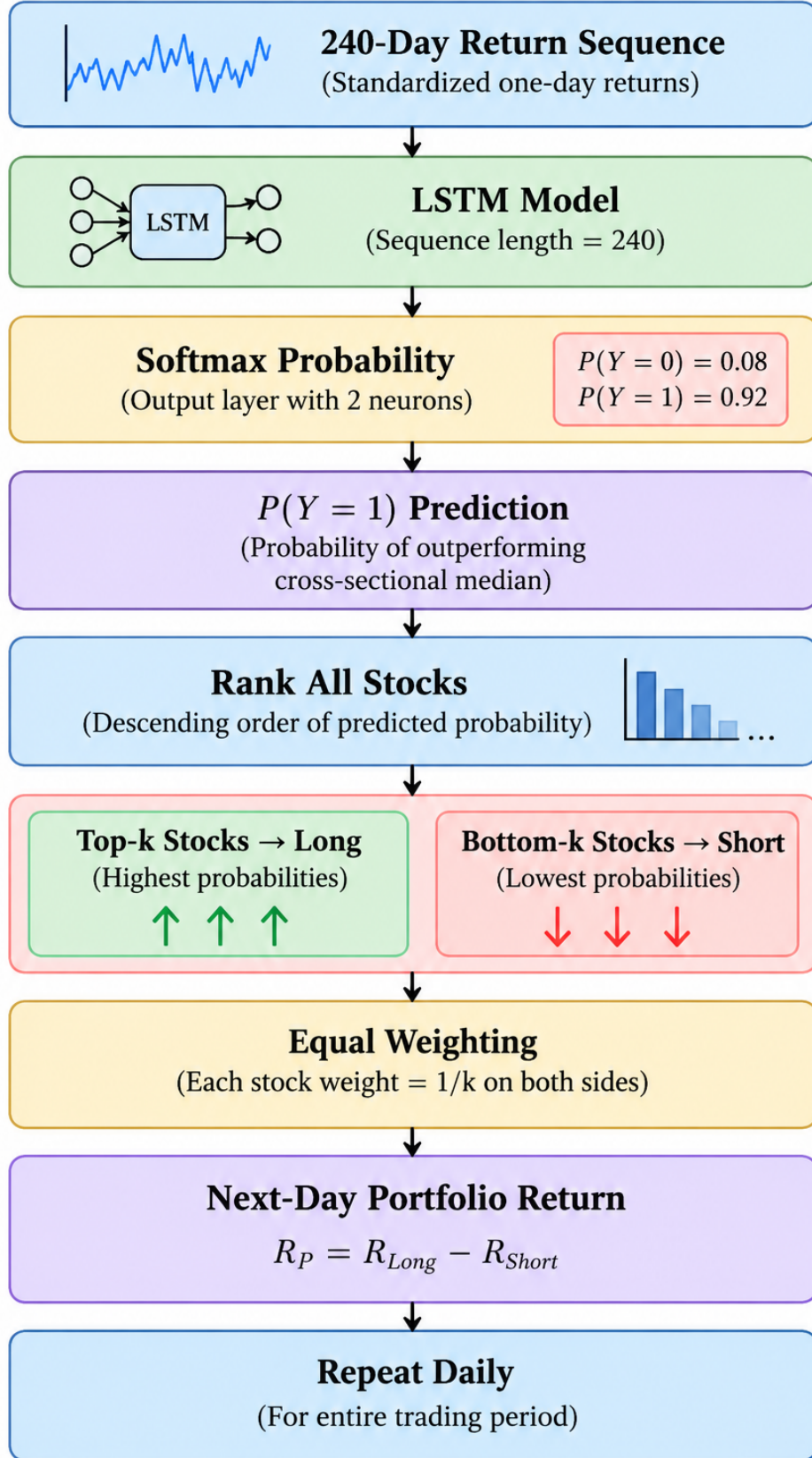


Figure 1: Overall forecasting and trading pipeline. A 240-day return sequence is provided to the LSTM model, which predicts the probability of a stock outperforming the cross-sectional median return on the next trading day. Stocks are ranked according to predicted probabilities, the top- k stocks are held long, the bottom- k stocks are held short, and portfolio returns are computed using an equal-weighted dollar-neutral strategy.